UFCEP6-0-3 · ADVANCED INFORMATION SYSTEMS DEVELOPMENT

# Hospital Patient Referral, Treatment & Administration System

## The second release: what we changed, why, and what it cost

### Three questions this deck answers

1. What did the second release actually change? — five changes, each with before and after
2. What did the feedback ask for, and what did we do with it? — fourteen items, twelve implemented
3. How good was our plan, really? — planned against actual, and what we would do differently

**Group: max · finn · henry · peter**

**Second release · 22 October 2026**

# From a model that deploys, to a service that is governed, persisted and evidenced

**The initial release proved the process could be drawn and executed. The second release proves it can be controlled, stored and tested.**

**4**
BPMN models
S1, S2, S3, O1

**7**
process definitions
deployed as one set

**69**
Camunda Forms
bound to user tasks

**18**
acceptance cases
in the harness

### Three capabilities that did not exist in September

- Role-based task routing — authorisation is on, and two roles see different inboxes
- A database — business records and the audit trail survive a restart
- A re-runnable acceptance harness — 18 cases in one command, evidence retained

### And three we still cannot claim

- Live integration with an external provider — all four are simulated
- Management reporting output — the monthly timer has not elapsed
- A tested availability or downtime procedure — no procedure exists to test

# What the initial release delivered, and the five gaps we already knew about

**The second release was scoped against these five, not against a wish list.**

## Delivered at the initial release (28 September)

- Operational model O1 deployed and executed on a real Camunda 8 cluster
- 68 Camunda Forms generated from the models and bound to their user tasks
- 18 acceptance cases executed against the deployed model — 18 passed, 0 incidents
- 41 worker unit tests; 0 moddle warnings; 0 Camunda 8 lint reports
- Defect log: 16 categories in the Sprint 1/2 files, 4 found only at deployment

## The five gaps, stated then and not hidden

- 1 Enforced role-based access — candidate groups expressed intent, the engine ignored membership
- 2 Immutable audit records — written to an in-memory map
- 3 Management reporting output — the monthly timer had not elapsed
- 4 Live external integration — scheduling, correspondence, clinical and payment were simulated
- 5 Availability / downtime — no procedure existed to test

**October status: gap 1 closed with evidence. Gap 2 partly closed. Gaps 3, 4 and 5 remain open and are stated on page 14.**

# Fourteen points of feedback, five themes, and two we did not implement

**Every point is traced to a decision, an action and an artefact. The two we did not implement are on this deck on purpose.**

| | | | | |
|---|---|---|---|---|
| A | **Execution gap in the model** | The booking path did not show what happens when the external service is unavailable. | 1 item | P-01 |
| B | **Evidence quality** | Criteria were narrative, claims were unsourced, and there was no before/after. | 4 items | P-02 04 06 08 |
| C | **Model integrity** | Pool names did not match the process names inside them. | 1 item | P-07 |
| D | **Governance and traceability** | No review rule, no i* timeline, no task id in the trace, no record of unimplemented actions. | 4 items | P-03 09 12 14 |
| E | **Not deliverable this release** | Urgency rules and an automated RBAC regression suite. | 2 items | P-10 11 |

**12 of 14 implemented · 2 not implemented, with the reason and the residual risk recorded · the two are named on the next page and justified on page 14**

### P-10 · urgency classification rules — not implemented

– **Why it stopped here**

– The case study says the hospital has not agreed a complete set of urgency rules, and the module team confirmed none would be issued before hand-in.

– **What we did instead**

– Urgency is modelled as a soft goal, not as a rule; the urgent-enquiry path stays discretionary.

– **Residual risk**

– Two patients with the same presentation could be triaged differently. An urgency-rules decision is a second-release Must.

### P-11 · automated RBAC regression — not implemented

– **Why it stopped here**

– Sprint 4 had no capacity left once the deployment defects and both evaluations were re-scoped.

– **What we did instead**

– RBAC is evidenced by a live two-role check (0 against 785, 0 failures) and by design inspection.

– **Residual risk**

– A regression could silently reopen a role. It is the one gap we have fixed but cannot auto-prove; a regression suite is a second-release Should.

# Feedback to decision to action to evidence to outcome — all fourteen rows

## Two rows are not implemented. They are marked, not omitted.

| ID | Feedback | Decision | Evidence | Outcome |
|---|---|---|---|---|
| P-01 | No path when the external service is unavailable | Add a pending-booking path with a recorded retry | T_KeepPending · TC-11 | Implemented |
| P-02 | Criteria must be measurable | Rewrite with a measurement method; keep the original wording | 28 revised criteria · Task 04 | Implemented |
| P-03 | No review rule in the repository | Protect main; record contribution evidence | git log · contribution matrix | Implemented |
| P-04 | Claims unsupported by the case study | Add a source column to assumptions and re-check requirements | assumption-register.md | Implemented |
| P-05 | SR model must be defensible branch by branch | Complete SR coverage; urgency as a soft goal | SR v1.0 · 61 elements | Implemented |
| P-06 | Prove the criteria are measurable | Measurement method plus a coverage map | 28 criteria · 18 cases | Implemented |
| P-07 | Pool names do not match process names | Rebuild as one model family, single participant register | 4 models · validator clean | Implemented |
| P-08 | Show the original defects are gone | Publish a before/after validation comparison | Appendix B · validation logs | Implemented |
| P-09 | i* must not be claimed to have informed the design | State the timeline wherever i* is used | Task 02 / 03 wording | Implemented |
| P-10 | Agree the urgency classification rules | Ask the module team for the rule | none — no rule issued | **Not implemented** |
| P-11 | Add automated RBAC regression tests | Automate role and audit checks | none — design inspection only | **Not implemented** |
| P-12 | Put the operational task id in the trace row | Combine the strategic and operational columns | Task 03 v2 | Implemented |
| P-13 | Give a recommendation against each failure | Add a recommendation per failure | Task 04 §6 | Implemented |
| P-14 | Record unimplemented actions with a reason | Record reason and outstanding implication | P-10, P-11 | Implemented |

P-01 to P-14 are this deck's own running numbers. The portfolio's feedback record numbers the same points in its own scheme; the substance, not the identifier, is what matches.

# Change 1 — two tasks were secretly parallel splits

**BPMN reads two unconditional outgoing flows as a parallel split. So every refund case both refunded the payment and retained it.**

**BEFORE · initial release**




no gateway → both branches always run

**AFTER · second release**




explicit gateway → exactly one branch

**3**
instances of the same defect found in the model family

**2**
implicit AND-splits replaced by explicit gateways

**0**
structural findings across all four models

**4**
models rebuilt on one participant register

| Defect | Model and element | Before | After |
|---|---|---|---|
| DEF-17 | S3 T_RefundDecision | two unconditional outgoing flows → refunded and retained at once | G_RefundDecision, refund branch conditional, retain is the default |
| DEF-18 | O1 / S3 T_AuthoriseUrgentTreatment | two unconditional flows → implicit AND-split | G_UrgentTreatmentAuthorised, an explicit parallel gateway |
| DEF-19 | S1 G_Accepted | branched on an accepted boolean no form collected → never left the default | branches on the controlled referralDecision, collected by a select |

# Change 2 — the forms collected less than the process already needed

**The models were correct. The forms in front of them were not: gateways compared values that no field could produce.**

**69**
Camunda Forms
68 rewritten, 1 new

**10**
control types
corrected

**41**
controls added for data
the process needed

**115**
validations added to
existing fields

### A value was typed, not chosen

– Four gateways compared exact strings, but the field was a free text box.
– referralDecision — accepted / rejected / information required / redirected
– referralUrgency — urgent / routine
– enquiryType — administrative / financial / clinical
– paymentStatus — four controlled values
– All four are select controls now, and all four are required.

### A number was a tick box

– The 7-day letter rule and the 1-month and 3-month escalations compare numbers.
– letterDelayDays — was checkbox, now number, 0 to 365
– letterAgeMonths — was checkbox, now number, 0 to 120
– A tick box can only ever be true or false, so the comparison could never be satisfied honestly.

### Three inputs no form collected

– Each fell back to a default chosen by code, not by a person.
– documentationCheckResult — now in form-check-referral-documents
– letterRecipients — now in form-clinic-letter-admin-check
– refundAuthorised — now in form-refund-decision
– 0 worker inputs remain with no form source.

# Change 3 — every role used to see every task; now they do not

**Two settings in the shipped engine configuration meant authorisation was never evaluated at all.**

**BEFORE · initial release**

authorizations.enabled = false

unprotected-api = true

**Every authenticated user saw every task. All ten roles had an identical inbox.**

**AFTER · second release**

authorizations.enabled = true

unprotected-api = false

**10 groups, 10 users, 99 candidate-group references read out of the models themselves.**

**0**

**tasks visible to a user with no role**

norole.demo · HTTP 200

**785**

**tasks visible to medical-secretaries**

secretary.demo · HTTP 200

**0**

**enforcement failures recorded**

re-checked 15 October 2026

– The grants are deliberately identical across all ten groups. What separates the roles is the candidate group on the task, not the permission set — which is exactly what 0 against 785 demonstrates.

| Group | Tasks routed | Group | Tasks routed |
|---|---|---|---|
| medical-secretaries | 6 | call-handling-team | 6 |
| consultant-team | 19 | finance-team | 6 |
| clinical-nurse-specialist-team | 2 | patient-pathway-coordinators | 0 |
| outpatient-bookings-team | 9 | administrative-management | 0 |
| treatment-bookings-team | 4 | hospital-management | 1 |

Full per-role extract, with its source hash: 05_Test_Evidence/rbac/rbac-visibility-evidence.pdf · recompute with 09_Tests_and_Tools/verify-rbac.py

# Change 4 — the records lived in maps; now they live in a database

**Two separate JVM processes. The second shares nothing with the first except the database file — so anything it reads back survived a restart.**

**H2**
embedded database
plain JDBC, no ORM

**4**
tables created on
first use

**7 / 7**
persistence checks pass
across two processes

**41**
unit tests still pass
unchanged behaviour

## The four tables

- record — business records keyed by episode, payload as JSON so new fields need no migration
- appointment_index — de-duplication now enforced by a primary key
- attempt_counter — incremented in one statement, so concurrent retries cannot lose a count
- audit_log — append-only, instant set at write time, no update or delete on the API

## What the reading process proved

- Referral REF-PERSIST-1 survived with its patient id — PASS
- Appointment APPT-PERSIST-1 survived — PASS
- The persisted payment still trips the duplicate-payment rule — PASS
- The appointment index still recognises the booking — PASS
- The attempt counter still reads 2; the audit log kept both entries — PASS
- The audit log still exposes no update or delete operation — PASS

**Stated limit: in this sandbox the H2 file lock cannot be acquired, so a second process opens the file read-only and only one process may write at a time. That is an environment restriction, not a property of the design — and it is why the proof is a write-then-exit-then-read, not a live two-writer test.**

# One run, five changes: deploy, start, two roles, bad input, persistence, evidence

**Live where the cluster is up; recorded backup cued on the desktop.**

| | | | |
|---|---|---|---|
| 1 | **Deploy** | node 09_Tests_and_Tools/deploy-all.mjs | 7 definitions and 69 forms in one command. |
| 2 | **Start** | Start Patient referral from Operate | An instance appears in Operate. |
| 3 | **Role A** | Tasklist as secretary.demo | The Medical Secretaries inbox. |
| 4 | **Role B** | Same Tasklist as finance.demo | Different inbox. RBAC is live, not designed. |
| 5 | **Bad input** | Open a form, submit it empty | Required and format validation stop it. |
| 6 | **Persist** | run-1-write, close, then run-2-read | 7 of 7, across two JVMs. |
| 7 | **Evidence** | Open the acceptance run log | The run log, dated, with its case count. |

**If the cluster does not start: play the recording, keep the log and screenshots on the slide. We do not gamble live.**

# Change 5 — the acceptance run, including the one that failed

**A criterion that has never failed has never been tested. We keep the failed run in the record.**

| Run | Date | Definitions | Cases | Incidents | Status |
|---|---|---|---|---|---|
| Initial release (tag sr2) | 26 September 2026 | 6 | 18 / 18 | 0 | Baseline — the run behind the initial release |
| Second-release rerun | 15 October 2026 | 7 | 7 cases hit UNHANDLED_ERROR_EVENT | 4 | Diagnosed and fixed |
| Second-release verification | October 2026 | 7 | per change: validator clean, 41 unit tests, 7/7 persistence, 0 vs 785 RBAC | — | Verified change by change, not re-run in full |

### What the failed run told us

- Three tasks raised the same error: T_NotifyReferralOutcome, T_SendAppointmentLetter, T_RedirectReferral.
- Seven cases hit it before the run died with ECONNRESET.
- Cause: the worker fleet was not subscribed when the run started, so the job was never picked up.
- All three are outbound messages, so it was one registration gap, not three separate defects.
- Fixed; the same cases pass when the fleet is up first.

### Why we keep it, and what it costs us

- The rubric asks for validation evidence, not for a clean history.
- The 18-case run belongs to the initial release and stays visible as its baseline.
- We did not complete a full re-run on the final build. Each change is verified by its own check instead — a narrower claim than a full run, and we say so.
- The 15 October run is retained as evidence rather than re-run until it passed.

**Three rows, three different strengths of claim. Only the first is a full end-to-end run. The third is proportionate to what changed — and we would rather show four targeted checks than one run we did not complete.**

# What the second release changed, in nine numbers

**Every value on the right is checkable in the repository; the left column is what the September review saw.**

| | September (initial release) | October (second release) | Where to check |
|---|---|---|---|
| Model family | 3 fragments, inconsistent pool names | 4 models, one participant register | 02_BPMN_Models |
| Process definitions | 6 | 7 | deploy-all.mjs |
| Camunda Forms | 68 | 69 | 03_Forms |
| Form control types | 2 (text, checkbox) | 6, with 115 validations | form-field-change-log.md |
| Implicit parallel splits | 2 undetected | 0, both given explicit gateways | defect-log.md DEF-17, DEF-18 |
| Role-based access | off — all roles, one inbox | on — 0 vs 785 tasks, 0 failures | rbac/task-visibility.json |
| Persistence | in-memory maps | H2, 4 tables, 7/7 across processes | persistence/run-2-read.txt |
| Verification | 18 acceptance cases, 6 definitions | 18 cases plus validator, 41 unit tests, RBAC and persistence checks | 05_Test_Evidence |
| Failed-run history | not recorded | kept, with cause and fix | acceptance-rerun.log |

# Planned against actual: where 196 planned hours met 128 nominal hours

**Sprint 3 was planned at 116 hours against 80 nominal. Sprint 4 at 80 against 48. Both overran on the metric we set ourselves.**

| Sprint | Planned | Nominal capacity | Over nominal | Delivered |
|---|---|---|---|---|
| Sprint 3 (13–19 Oct) | 116 h | 80 h | +36 h | v0.9 — 3 of 6 review goals met |
| Sprint 4 (20–22 Oct) | 80 h | 48 h | +32 h | v1.0 — 5 changes, both evaluations |

### Cause 1 — an abstraction level agreed late

– Sprint 3 spent its first half deciding how many strategic models there should be.

– Consequence: the SR model, the alignment narrative and the acceptance evidence all missed SR3.

– v0.9 was presented with 3 of 6 goals met.

### Cause 2 — deployment-only defects

– Three XSD sequence violations and one form-binding defect appeared only at deployment.

– They were not in the plan, because the plan assumed the Modeler's parse was sufficient.

– Consequence: Sprint 4 lost capacity to defects nobody had estimated.

### Consequence — what we dropped

– P-11, an automated RBAC regression suite: no capacity left after the re-scope.

– P-10, urgency rules: blocked, not dropped — no rule was issued to us.

– Final integration landed in the last two days of Sprint 4.

**Net position: 196 hours were planned against 128 hours nominal — a 53% over-commitment made at planning time, not at delivery.**

Sprint 4 was planned at 167% of nominal capacity, so the final integration landing in the last two days was a consequence of the plan we wrote, not a surprise that arrived.

# Three lessons we would carry forward, and six things we still do not claim

**The right-hand column is the one we want to be marked on.**

## Carry forward

– **Rehearse integration at mid-sprint, not at the end**

– Both sprints ended with integration inside the last two days, and both times the defects it found were the expensive ones.

– **Build the test harness before the exception paths are specified**

– The 18 cases were written after the model changed, so every model change cost a test change.

– **Treat documentation as scheduled work, not as slack**

– The traceability table and the evaluations were the two items that slipped first and hurt most.

– **Freeze, tag and re-run before the last day, not on it**

– The final acceptance run, the tag and the re-exported figures were still open on the morning of hand-in. They should have been planned work with an owner and a date.

| What we do not claim | Why |
|---|---|
| Enforced RBAC with an automated regression suite | P-11 dropped for capacity; verified by inspection |
| Audit records immutable at database level | Append-only API and table; no DB grants |
| Management reporting output | The monthly timer has not elapsed |
| Live integration with any external provider | All four services are simulated |
| A tested availability or downtime procedure | No procedure exists to test |
| Two processes driven end to end | Enquiry and reporting deploy but were not driven |

**Every row in the right-hand table is a claim we could have made and chose not to. The one that matters most is the first: role-based access now works, and we still cannot prove automatically that it keeps working.**

# What we claim, what we do not, and where we would take it next

**The second release is scoped against the feedback, evidenced against the baseline, and honest about what it still cannot do.**

### We claim

– Five changes, each traceable to a feedback item and to a repository artefact

– Role-based routing and a database that survive a restart, both with evidence

– A re-runnable acceptance harness that records a failed run as well as a passing one

### We do not claim

– Live external integration, management reporting output, or a tested downtime procedure

– Audit immutability below the application layer

– An automated RBAC regression suite

### Next release, in priority order

– 1 RBAC regression suite — the only gap we have already fixed but cannot auto-prove

– 2 Replace the four simulated services with one real provider

– 3 Point the database at a server instance; the code is already plain JDBC

## Discussion we would welcome

– Is an acceptance harness that has only ever passed worth more, or less, than one that has failed and been fixed?

– Should role-based access be evidenced by automated regression, or is a design inspection sufficient at this stage?

– How far should external integration go before the simulated boundary stops being defensible?

### References

Camunda (2024) Camunda 8 Documentation. · Dalpiaz, F., Franch, X. and Horkoff, J. (2016) 'iStar 2.0 language guide', arXiv:1605.07767. · H2 Database Engine (2024) H2 Documentation. · Horkoff, J. and Yu, E. (2016) 'Interactive goal model analysis for early requirements engineering', Requirements Engineering, 21(1), pp. 29–61. · Object Management Group (2011) Business Process Model and Notation (BPMN) v2.0. · Recker, J. (2010) 'Opportunities and constraints: the current struggle with BPMN', Business Process Management Journal, 16(1), pp. 181–201. · Schwaber, K. and Sutherland, J. (2020) The Scrum Guide. · UWE Bristol (2026) Case Study: Hospital Patient Referral, Treatment and Administration System.

**Thank you — questions welcome. Four backup pages follow: A1 the requirements trace, A2 the acceptance criteria, A3 the model inventory, A4 the demo fallback.**

# 44 requirements: 28 fully supported, 12 partly, 2 not supported, 2 out of scope

**Full table in UFCEP4-0-3 Task 03. Shown here only if asked.**

| Verdict | Count | Meaning | Example |
|---|---|---|---|
| Fully supported | 28 | Traced to a strategic activity, an i* element and a deployed artefact | REQ-09 the 14-day telephone rule — gateway, form field, TC-06 |
| Partly supported | 12 | Traced, but one layer is missing or evidenced by inspection only | REQ-37 audit immutability — append-only API, no DB grants |
| Not supported | 2 | In scope, no implementation | Management reporting output — timer has not elapsed |
| Out of scope | 2 | Deliberately outside the first two releases | Live external integration |

# 20 original criteria rewritten as 28 measurable ones, in four families

**The four worked examples below are the ones we usually get asked about.**

| 11 | 9 | 5 | 3 |
|---|---|---|---|
| normal path | exception path | business rule | non-functional |

## Why the original twenty could not fail

- They described activities, not outcomes — "the referral is handled".
- Only two carried a threshold, so eighteen had no way to be false.
- Nine exception requirements had no criterion at all.
- Non-functional needs were one unmeasurable line.

## AC-R-12 — the one worth memorising

- Given a payment the provider took but did not confirm,
- when no confirmation is returned,
- then the transaction is marked for investigation
- and no second payment request is raised.
- **Traces to REQ-23 · BR-06 · TC-16**

# Four models, seven definitions, sixty-nine forms — the counts behind every slide

**Generated from the .bpmn and .form files on 20 October 2026; see model-inventory.md.**

| Model | Flow nodes | Lanes | Message flows | Process definition(s) |
|---|---|---|---|---|
| S1 Strategic landscape | 25 | — | 15 | PR_Landscape |
| S2 Referral to authorisation | 37 | 7 | 9 | PR_ReferralToAuthorisation |
| S3 Treatment to aftercare | 79 | 7 + 3 + 1 | 13 | PR_TreatmentToAftercare, PR_EnquiryHandling, PR_ManagementReporting |
| O1 Operational merged | 60 | 8 | 15 | PR_Operational_Merged, PR_ReferringOrganisation |
| **Total** | **201** | | **52** | **7 definitions · 69 Camunda Forms** |

**Not driven end to end by the acceptance run: PR_EnquiryHandling and PR_ManagementReporting. Both deploy and start; neither was exercised by the 18 cases.**

# If the cluster will not start, and where each part of the run lives

**The recording is a fallback, not a substitute: the log and the screenshots stay on the slides either way.**

| Step | If live | If not live | Artefact it produces |
|---|---|---|---|
| Deploy | deploy-all.mjs | recorded clip 00:00–0:40 | 7 definitions, 69 forms in Operate |
| Start | Start Patient referral | recorded clip 00:40–1:05 | one running instance |
| RBAC | two logins, two inboxes | two screenshots (to be re-taken) | task-visibility.json |
| Validation | empty form submit | recorded clip 01:20–1:40 | form error message |
| Persistence | write, exit, read | run-2-read.txt on screen | 7/7 checks |
| Evidence | open the run log | acceptance-run.log on screen | run date, 18 cases, incidents |